

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1– Analytical Problem Solving

Course Code:	ITA0512	Course Title:	Computer Vision
CO Assessed:	CO2 – Imitation (BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	10% of CIA	Total Marks:	100
CO1 Statement:	Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)		
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Section / Batch:	B. Tech AI & ML	Date:	06 - 08 - 2026

Code of Conduct

I, S. Susannal Devapriyam (192525398) that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S. Susannal Devapriyam

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Criteria	Marks
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	Problem Understanding	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	Method Selection	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	Solution Process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	Accuracy of Calculation	
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	Interpretation of Results	

Questions :

19. Edge Detection (Vertical Sobel)

Apply vertical Sobel:

Matrix:

- (a) Interpret edge detection objective.
- (b) Justify operator choice.
- (c) Apply step-by-step.
- (d) Compute center value.
- (e) Interpret edge direction.

20. Prewitt Edge Detection

- (a) Interpret Prewitt operator purpose.
- (b) Justify use over Sobel.
- (c) Apply kernel:
 $\begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{bmatrix}$
- (d) Compute output.
- (e) Interpret result.

21. Region Splitting

Pixel set: [40, 45, 100, 105]

- (a) Interpret segmentation problem.
- (b) Justify splitting method.
- (c) Apply threshold 80.
- (d) Compute regions.
- (e) Interpret separation.

19. Edge Detection (Vertical Sobel)

Let the Given Image Matrix Be :

$$A = \begin{bmatrix} 10 & 40 & 70 \\ 20 & 50 & 80 \\ 30 & 60 & 90 \end{bmatrix}$$

Vertical Sobel Kernel

$$G_x = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

(a) Interpret Edge Detection Objective

The objective of edge detection is to identify object boundaries by detecting sudden changes in pixel intensity. It helps in image segmentation and feature extraction.

(b) Justify Operator Choice

The Vertical Sobel operator is selected because it detects **vertical edges** by measuring intensity changes in the horizontal direction. It also reduces noise because the center row has higher weights.

(c) Apply Step-by-Step

Multiply corresponding elements.

$$(-1 \times 10) + (-2 \times 40) + (-1 \times 70) + (0 \times 20) + (1 \times 30) + (0 \times 50) + (2 \times 60) + (0 \times 80) + (1 \times 90)$$

(d) Compute Center Value

$$= (-10) + 0 + 30 - 80 + 0 + 120 - 70 + 0 + 90 = 80$$

Center Pixel = 80

Final Output Matrix:

[10 20 30]

[40 80 60]

[70 80 90]

(e) Interpret Edge Direction

The positive output indicates a strong **vertical edge**. A higher magnitude represents a sharper intensity change from left to right.

20. Prewitt Edge Detection

Let the Given Image Matrix Be :

A = [10 20 30]

[40 50 60]

[70 80 90]

Prewitt Vertical Kernel

P = [-1 0 1]

[-1 0 1]

[-1 0 1]

(a) Interpret Prewitt Operator Purpose

The Prewitt operator detects edges by measuring changes in intensity between neighboring pixels. It is mainly used to identify object boundaries in an image.

(b) Justify Use over Sobel

The Prewitt operator uses equal weights, making it computationally simpler than the Sobel operator. It is suitable for applications requiring faster edge detection.

(c) Apply Kernel

Multiply corresponding elements.

$$(-1 \times 10) + (0 \times 20) + (1 \times 30) + (-1 \times 40) + (0 \times 50) + (1 \times 60) + (-1 \times 70) + (0 \times 80) + (1 \times 90)$$

(d) Compute Output

$$= -10 + 0 + 30 - 40 + 0 + 60 - 70 + 0 + 90 = 60$$

Output = 60

Final Output Matrix:

$$\begin{bmatrix} 10 & 20 & 30 \end{bmatrix}$$

$$\begin{bmatrix} 40 & \underline{60} & 60 \end{bmatrix}$$

$$\begin{bmatrix} 70 & 80 & 90 \end{bmatrix}$$

(e) Interpret Result

The positive output indicates the presence of a **vertical edge**. The magnitude of **60** shows a moderate intensity variation in the image.

21. Region Splitting

(a) Interpret the Segmentation Problem

Image segmentation is the process of dividing an image into multiple regions based on similar characteristics such as intensity. The given pixel values must be separated into different regions using the specified threshold value.

(b) Justify the Splitting Method

Region splitting is chosen because it separates pixels according to a threshold intensity. It is simple, computationally efficient, and groups pixels with similar gray-level values into different regions.

(c) Apply Threshold = 80

Threshold Rule:

- Pixel value $< 80 \rightarrow$ Region 1
- Pixel value $\geq 80 \rightarrow$ Region 2

(d) Compute the Regions

Given Pixels:

40 45 100 105

Applying threshold 80:

Region 1

[40 45]

Region 2

[100 105]

(e) Interpret the Separation

The threshold successfully separates low-intensity pixels from high-intensity pixels. Region 1 contains darker pixels, while Region 2 contains brighter pixels. This segmentation helps identify different objects or areas within an image.